Bayesian Discrete-Time Survival Modeling with Sensitivity Analysis for Informative Censoring: An Application to ICU Mortality

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1. Introduction

Accurate mortality risk profiling in the Intensive Care Unit (ICU) is essential for clinical decision-making, resource allocation. Essentially all survival models rely on the independence censoring assumption, that is, the censoring mechanism (e.g., hospital discharge) carries no information about the underlying survival process beyond observed covariates (Kalbfleisch and Prentice, 2003). In the ICU, this assumption is immediately suspect. Patient discharge is a purposeful clinical decision, exactly dependent on patient health status, creating a dependence structure (of censoring C_i and failure time T_i), violating the standard assumption, and biases hazard estimates and covariate effects.

Tsiatis (1975) Establishes a fundamental identifiability issue: from the observed data (y_i, δ_i) alone, marginals of T_i and C_i cannot be separately identified without structural assumptions on their joint distribution. Diagnostics too, cannot distinguish informative from non-informative censoring, motivating the need for sensitivity analyses.

This paper addresses the central question: *how robust are Bayesian posterior inferences about ICU mortaliy risk fators to departures from the independent censoring assumption?* We make three contributions within the discrete-time framework. First, we formalize the independent censoring assumption in the missing data taxonomy, deriving explicit bias formulas for hazard estimation and covariate effects (Section 3.1, Appendix A). Second, we develop a sensivity analysis following (Scharfstein et al., 1999) by introducing γ that indexes the departure from MAR, identifying *tipping points* where clinical conclusions are changed (Section 3.6). Third, we compare this with a shared frailty model (Section 3.5) and demonstrate through both simulation (Appendix C) and application to 57,694 MIMIC-III patients (Section 5) that frailty is weakly identified and that sensitivity analysis is the more principled approach to account for informative censoring in this context.

2. Data Structure & Structure

Let $i = 1, \ldots, N$ index unique ICU admissions from MIMIC-III. We define $T_i > 0$ as latent time to death, $C_i > 0$ as thhe censoring time (i.e., time of hospital discharge), and

$y_i = \min(T_i, C_i)$ as the observed follow-up time. The failure indicator is $\delta_i = \mathbb{I}(T_i \leq C_i)$, indicates whether the patient died in the hospital ($\delta_i = 1$) or was discharged alive ($\delta_i = 0$), in days. And $\mathbf{x}_i \in \mathbb{R}^P$ is a vector of static baseline covariates measured at ICU admission. The observed data for patient i is thus $(y_i, \delta_i, \mathbf{x}_i)$. The data covariate characteristics are described in more detail in Section 5, and Table 5.

2.0.1 Discretization.

We partition continuous time into discrete intervals of width Δ: The intervals are $[0, \Delta), [\Delta, 2\Delta), \ldots, [(K-1)\Delta, K\Delta)$ with $t = 1, \ldots, K$. The discrete failure time is $\tilde{T}_i = \lceil T_i/\Delta \rceil$ and similarly, $\tilde{C}_i = \lceil C_i/\Delta \rceil$. Our primary analysis uses daily discretization ($\Delta = 1$ day), over 30-day window ($K = 30$), motivated by the standard use of 30-day ICU outcome metric, the median follow-up of ≈ 6 days in MIMIC-III, so the window captures the clinically relevant risk period. We will proceed by dropping the tildes for notational simplicity, so T_i and C_i refer to the discretized times.

Our simulation study shows that covariate effect estimates are stable across $\Delta = \{1, 3, 7, 14\}$ (Appendix C). The choice of Δ involves a bias–variance trade-off: finer discretization provides a closer approximation to the underlying continuous-time process, but producing a larger dataset and more interval-specific parameters to estimate, which can lead to finite-sample instability.

2.0.2 Person-Period expansion.

Following (Allison, 1982), the data is reconstructed into a "person-period" format, so each patient i contributes one row per interval $t = 1, \ldots, K_i$ where $K_i = \min(T_i, C_i)$ is the last interval for which patient i is at risk. At time $t = K_i$, the binary outcome is $y_{i,K_i} = \mathbb{I}(\delta_i = 1)$, is either 1 if patient i dies, or 0 if censored at time $t = K_i$. These are absorbing states, so patients exit the risk set thereafter. Considering who is still under observation at each time t is critical to further analysis, as the risk set at time t is $\mathcal{R}_t = \{i : y_i \geq t\} = \{i : T_i \geq t, C_i \geq t\}$ and the censored set is $\mathcal{C}_t = \{i : C_i < t\}$.

3. Methodology

3.1 Standard Likelihood & Censoring Assumption

The discrete-time hazard for patient i at time t is $h_{i,t} = P(T_i = t \mid T_i \geq t, \mathbf{x}_i)$, with the corresponding survival function $S_i(t) = P(T_i > t \mid \mathbf{x}_i) = \prod_{s=1}^{t}(1 - h_{i,s})$. The full data likelihood involves the joint density of (y_i, δ_i) depending both on failure and censoring parameters $\boldsymbol{\theta}, \boldsymbol{\phi}$ respectively. (Kalbfleisch and Prentice, 2003, Section 3.2):

$$\mathcal{L}_i^{\text{full}}(\boldsymbol{\theta}, \boldsymbol{\phi}) = \left[f_T(y_i \mid \mathbf{x}_i)\, S_C(y_i \mid \mathbf{x}_i)\right]^{\delta_i} \left[S_T(y_i \mid \mathbf{x}_i)\, f_C(y_i \mid \mathbf{x}_i)\right]^{1-\delta_i} \tag{1}$$

Assumption: *[Independent Right-Censoring; K&P Section 2.4] $T_i \perp C_i \mid \mathbf{x}_i$: knowing the censoring time provides no additional information about survival beyond $\mathbf{x}_i$.*

Under Assumption 3.1, (1) factors so that inference about $\boldsymbol{\theta}$ proceeds without modeling censoring. In person-period form: $\mathcal{L}(\boldsymbol{\theta}) = \prod_{i=1}^{N} \prod_{t=1}^{K_i} h_{i,t}^{y_{i,t}} (1 - h_{i,t})^{1-y_{i,t}}$, where each (i, t) term is an independent Bernoulli trial. This corresponds to Missing at Random (MAR) in

Rubin (1976)'s taxonomy. Since time-varying information is likely critical to drive discharge decisions, and thus the censoring mechanism is likely informative (i.e., $T_i \not\perp C_i \mid \mathbf{x}_i$), the MAR assumption is likely violated in this context, motivating the methods below.

3.2 Consequences of Informative Censoring

When the independent censoring assumption fails, two consequences follow. First, systematic discharge of healthier patients inflates the observed hazard. Where $\lambda_t = |\mathcal{C}_t|/|\mathcal{R}_t|$ is the proportion of patients censored at t, and $\Delta_t = h_t^* - \hat{h}_t$ is the difference between the true marginal hazard h_t^* and the observed hazard $\hat{h}_t$. Then Appendix A:

$$\hat{h}_t \approx h_t^* + \frac{\lambda_t \Delta_t}{1-\lambda_t} \quad \text{where} \quad \hat{h}_t = \frac{\sum_{i \in \mathcal{R}_t} \mathbb{I}(T_i = t)}{|\mathcal{R}_t|} \tag{2}$$

This bias is positive whenever, λ_t and $\Delta_t > 0$ (i.e., censored patients are healthier). But Δ_t could be negative indicating censored patients are sicker, which would yield a negative bias term. This is revisited in Subsection 3.5.1.

Second, covariate information that inform early discharge and lower mortality risk (e.g., elective admission, lower comorbidity scores) will have their effects $\boldsymbol{\beta}$ confounded by the censoring process. (Scharfstein et al., 1999) formalizes this as a failure of the semiparametric estimation equations when MAR is violated, where the score function for $\boldsymbol{\beta}$ is biased proportional to the dependence between T_i and C_i given $\mathbf{x}_i$ (the selection bias). This is formalized in Appendix B, where the bias in the score function for $\boldsymbol{\beta}$ under informative censoring can be expressed as:

$$\text{Bias}(\alpha) \approx -\alpha\, \mathbb{E}\Big[\Lambda_0(C_i \mid \mathbf{x}_i)\, T_i\, S_{\text{full}}(\boldsymbol{\beta})\Big] \tag{3}$$

In the context of this project, the full-data score ($S_{\text{full}}(\boldsymbol{\beta})$) is Equation 1. This bias is driven by α (the severity of the MAR violation/selection) and the covariance (cumulative hazard) between the failure time T_i and the censoring $\Lambda_0(C_i \mid \mathbf{x}_i)$. Because α cannot be identified from the observed data alone, standard estimation fails motivating the sensitivity analysis and frailty model.

Though the frailty model sought to absorb the heterogeneity the drive censoring and failure, our demo simulation demonstrates this approach faces identifiability challenges. The primary model, Model A, is given below and compared against the frailty extension, Model B.

3.3 Logistic Hazard Model

The persion-period formulation recasts the discrete-time survival hazard as a first-order Markov chain on the state space $\{1, 0\}$ where 0 denotes "alive" and 1 denotes "dead" (an absorbing state). At each interval t, a patient who survives to that point ($y_{i,t-1} = 0$) either transitons to death ($y_{i,t} = 1$) or remains alive ($y_{i,t} = 0$). Conditioning on previous interval $y_{i,t-1} = 0$, requires that all preceding intervals are also zero. The discrete-time hazard precisely the one-step transition probability, which can be modeled as a Bernoulli GLMM with a logistic link.

3.4 Standard Model (Model A)

The primary analysis model specifies the hazard via a logistic link within interval-specific intercepts and shared covariate effects:

$$y_{i,t} \mid h_{i,t} \sim \text{Bernoulli}(h_{i,t}), \quad t = 1, \ldots, K_i,\ i = 1, \ldots, N, \tag{4}$$

$$\text{logit}(h_{i,t}) = \alpha_t + \mathbf{x}_i^\top \boldsymbol{\beta} \tag{5}$$

with priors:

$$\alpha_1 \sim \mathcal{N}(0, \sigma_0^2), \qquad \text{(anchor)} \tag{6}$$

$$\alpha_t \mid \alpha_{t-1} \sim \mathcal{N}(\alpha_{t-1}, \sigma_\alpha^2), \qquad t = 2, \ldots, K, \tag{7}$$

$$\sigma_\alpha \sim \text{Student-}t^+(\text{df=}3, 0, 1), \tag{8}$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}_P). \tag{9}$$

The first-order random walk prior on $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)^\top$ encodes the temporal correlaition and the belief that neighboring intervals have similar baseline log-odds of death. As the risk set shrinks at later intervals, this prior borrows strength from adjacent time points, regularizing α_t where data is sparse. The half Student-t prior on σ_α is weakly informative and avoids the boundary issues of conjugate Inverse-Gamma priors with small shape parameters (Gelman, 2006). We set $\sigma_0^2 = \tau^2 = 10$ as weakly informative defaults, allowing shrinkage of both α_t) and $\boldsymbol{\beta}$ towards zero when data is sparse, while still allowing for large effects when supported by the data. Let $\boldsymbol{\theta} = (\boldsymbol{\alpha}, \boldsymbol{\beta}) \in \mathbb{R}^{K+P}$. Appendix C guides justification here, showing well-identified and convergent under under HMC, with ESS in the thousands for all β_j and no divergences.

3.5 Frailty Extension (Model B)

To capture unobserved patient-level heterogeneity, we extend Model A with a Gaussain random effect b_i (frailty) for each patient i to account for the *within* patient correlation across intervals:

$$\text{logit}(h_{i,t}) = \alpha_t + \mathbf{x}_i^\top \boldsymbol{\beta} + b_i \tag{10}$$

with additional priors:

$$b_i \sim \mathcal{N}(0, \sigma_b^2), \qquad i = 1, \ldots, N, \tag{11}$$

$$\sigma_b \sim \text{Exp}(1). \tag{12}$$

the penalized complexity (PC) prior as a standard exponential, favoring smaller σ_b^2 (Simpson et al., 2017) which shrinks towards no heterogeneity $b_i = 0$ which is simply Model A. This is more principled than the Inverse-Gamma hyperprior for this setting, as it avoids both the small shape parameter placing excessive mass near zero and having heavy right tails.

3.5.1 Justification & Limitations.

To account for unobserved heterogeneity that may drive both censoring and failure, we introduce a patient-specific frailty term b_i in the linear predictor. By conditioning on b_i,

the model relaxes the strict independence assumption (3.1) to a strictly weaker condition (13), requiring only the augmented covariate set $(\mathbf{x}_i, b_i)$:

$$T_i \perp C_i \mid \mathbf{x}_i, b_i \tag{13}$$

An important caveat is that addition of b_i does not guarentee (13) holds, as the frailty is a misspecified proxy for the true unobserved confounder(s) that drive both censoring and failure. If b_i captures some (not all) of the relevant heterogeneity, then there will still be residual informative censoring bias. This can be though of as a partial correction to (2), where the bias is attenuated by the degree to which b_i captures heterogeneity. By individually estimating these extra N random effects, it allows for censoring of either improving ($b_i < 0$) or deteriorating ($b_i > 0$) patients, which would yield a positive/negative bias term in (2) respectively. However, the downside is that it significantly increases the dimensionality of the estimation problem, and thus identifiability becomes a major concern, plummeting ESS for σ_b^2 and b_i in practice.

3.6 Sensitivity Analysis via Exponential Tilting

The sensitivity analysis on Model A (only) and frailty model address complementatry aspects of informative censoring. The random effect b_i captures some heterogenity *parametrically*, assuming the Gaussian prior (11) and condition (13). The sensitivity analysis, by contrast, *nonparametrically* aims to index the departures from independent censoring without specification of a particular dependency structure.

Following Scharfstein et al. (1999) and the Bayesian treatment of Daniels and Hogan (2008), define γ as the log-odds ratio of death at time $T_i = t$ between patients censored at t and those remaining under observation at t:

$$\gamma = \log\left(\frac{\Pr(T_i = t \mid T_i \geq t, C_i = t, \mathbf{x}_i)}{\Pr(T_i = t \mid T_i \geq t, C_i > t, \mathbf{x}_i)}\right) \tag{14}$$

Here, $\gamma = 0$ corresponds to MAR, $\gamma < 0$ means censored patients have lower mortality risk (discharge reflects improvement), and $\gamma > 0$ means censored patients have higher mortality risk (discharge reflects transfer to hospice). γ is equivalent of the selection bias parameter "α" in (3), indexing the residual selection bias that cannot be captured by $\mathbf{x}_i$ alone. By sweeping γ over a grid, we can assess how sensitive our inferences are to departures from the MAR assumption, without parametric assumptions or identifiability issues. Note that we conduct the sensitivity analysis on Model A and do not condition on b_i, so γ indexes the *total* departure from MAR, rather than the *residual* departure beyond what the frailty captures.

3.6.1 Tilted Hazard & Likelihood.

The tilted hazard $\tilde{h}_{i,t} = P(T_i = t \mid T_i \geq t, \mathbf{x}_i, \gamma)$ adjusts the censored observations at the terminal interval:

$$\text{logit}(\tilde{h}_{i,t}) = \text{logit}(h_{i,t}) + \gamma \mathbb{I}(C_i = t, \delta_i = 0), \qquad \text{where } \delta_i = \mathbb{I}(T_i \leq C_i) \tag{15}$$

Equivalently, the mortality odds are multiplied by $\exp(\gamma)$ at the censoring interval. For uncensored observations and non-terminal intervals, $\tilde{h}_{i,t} = h_{i,t}$. The tilted likelihood is:

$$\tilde{\mathcal{L}}(\boldsymbol{\theta}; \gamma) = \prod_{i=1}^{N} \prod_{t=1}^{K_i} \tilde{h}_{i,t}^{y_{i,t}} (1 - \tilde{h}_{i,t})^{1-y_{i,t}} \tag{16}$$

This reweights the original likelihood, where their contribution of censored observations at their terminal interval is adjusted by γ. Since γ is fixed and the censoring indicator is observed, it is simply Model A with an offet at censoring time:

$$\text{logit}(\tilde{h}_{i,t}) = \alpha_t + \mathbf{x}_i^\top \boldsymbol{\beta} + o_{i,t} \tag{17}$$

with tilt term $o_{i,t} = \gamma \mathbb{I}(C_i = t, \delta_i = 0)$. Unlike σ_b^2, γ is swept over a grid rather than being estimated, making the analysis immune to identifiability issues. The *tipping point* γ_j^* for covariate j is the value within the 95% CrI for β_j first includes zero, quantifying the degree of MAR violation required to change clinical conclusions about covariate j.

3.7 Protocol.

Following Daniels and Hogan (2008):

1. Fit the standard model (Model A) with $\gamma = 0$ and record posterior summaries of $\boldsymbol{\beta}$, σ_α, and $\hat{S}(t \mid \mathbf{x})$.
2. Select a grid $\gamma \in \{-2.0, -1.5, -1.0, -0.5, -0.25, 0, 0.25, 0.5, 1.0, 1.5, 2.0\}$, where negative values correspond to the clinically plausible direction (healthier patients discharged earlier).
3. For each γ, refit Model A with the tilted likelihood (16).
4. Report posterior means and 95% credible intervals for each β_j as a function of γ (sensitivity plot).
5. Identify the tipping point γ_j^* for each covariate: the value at which the 95% credible interval for β_j first includes zero.

If conclusions are robust across the grid of γ, the standard analysis assuming MAR is sufficient, despite the potential for informative censoring. If conclusions change at modest values of $|\gamma|$, this quantifies the fragility of the findings.

4. Computation

We initially developed a Pólya-Gamma (PG) Gibbs sampler (Polson et al., 2013) for the logistic hazard model to leverage conjugacy for efficient posterior computation. However, the centered parameterization of the frailty model exhibited Neal's funnel geometry (Neal, 2003), for small σ_b^2 the frailties b_i mixed well, but for larger values the sampler failed to explore the posterior due to the strong posterior correlation, yielding very low ESS. This motivated the switch to Hamiltonian Monte Carlo (HMC) via Stan (Stan Development Team, 2024b).

4.1 Hamiltonian Monte Carlo via Stan

We implement all models in Stan, which uses the No-U-Turn Sampler (NUTS; Hoffman and Gelman, 2014), a self-tuning variant of HMC, well-suited for high-dimensional posteriors.

The key improvement is a *non-centered* reparameterization of Model B: $b_i^{\text{raw}} \sim \mathcal{N}(0,1)$, $b_i = \sigma_b \cdot b_i^{\text{raw}}$, decoupling the sampling of b_i and σ_b. Combined with the PC prior on σ_b, this yields substantially better mixing for σ_b and b_i. For both models, we run 4 chains with 500 warmup and 2,000 sampling iterations, yielding 8,000 posterior samples in CmdStanR (**?**) due to `rstan`'s extra overhead.

For the sensitivity sweep, full posterior sampling with HMC for the 7 total γ values would require 11 separate HMC runs, and given the massive scale of the person-period dataset ($R = 509,944$ rows), this is computationally prohibitive. We use Stan's Automatic Differentiation Variational Inference (ADVI) (full-rank) (Stan Development Team, 2024a), which approximates the posterior with a tractable family and iteratively optimizes the parameters to minimize the KL divergence to the true posterior. While ADVI underestimates heavy-tail uncertainty, it is sufficient for identifying the tipping points where the 95% credible intervals first cross zero and is orders of magnitude faster than HMC. The HMC results for the standard model (Model A) at $\gamma = 0$ serve as the gold standard, ADVI results across the γ grid provide a computationally efficient sensitivity analysis.

5. Data Analysis: Application to MIMIC-III

5.1 Cohort & Data Structure

We apply our framework to MIMIC-III (Johnson et al., 2016), a freely accessible critical care database from the Beth Israel Deaconess Medical Center (2001–2012). After restricting to adult first-admissions with complete covariate information, the analytic cohort comprises $N = 57,694$ patients. Using daily discretizaiton over a 30-day observation window ($\Delta = 1$ day, $K = 30$), patient outcomes decompose into: 5,443 deaths (9.4%), 49,660 discharges (86.1% clinically informative censoring under 3.1), and 2,591 surviving patients administratively censored at 30 days (4.5%). The person-period expands to $R = 509,944$ rows, with a median of 5.9 intervals per patient (IQR: 3.5–10.6). The front-loading discharge pattern is consistent with the typical short ICU stay, and underscores the potential for informative censoring bias in the standard analysis. See Appendix **??** for detailed cohort tables.

5.2 Covariate Matrix

The design matrix $\mathbf{X} \in \mathbb{R}^{K \times P}$ where $P = 17$ includes baseline covariates measured at ICU admission (Table 5). Nine of the covariates mirror those used in the simulation study: standardized age, sex, emergency admission, ICD-coded flags for sepsis, heart- and respiratory failure, and acute kidney injury, Glasgow Coma Scale impairment (GCS ¡ 12), vasopressor used within the first 24 hours. Eight additional clinical predictors extended the covariate set to include: ICD-flags for pneumonia, diabetes, chronic liver disease, malignancy, and COPD, along with indicators for mechanical ventilation, renal replacement therapy, and positive blood cultures. All covariates are binary except for age (standardized).

Table 5 presents these baseline covariates stratifed by outcome. Some notable patterns include: patients who died are substantially older (approximate mean age 70 vs. 61 for censored), more likely to have been admitted emergently, and markedly higher rates of sepsis, repiratory failure, and vasopressor use. The discharged-alive group, by contrast,

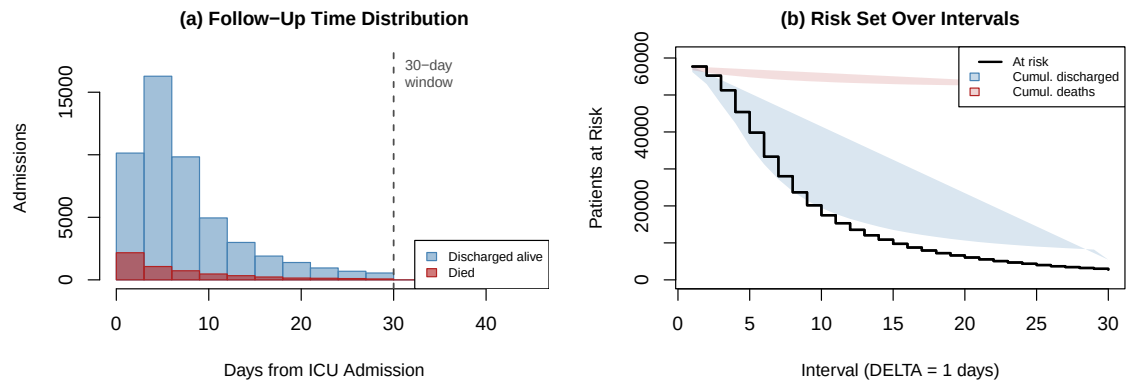

Figure 1: Empirical follow-up distribution and risk set evolution for the MIMIC-III cohort ($N = 57,694; \Delta = 1$ day, $K = 30$). **(a)** Histogram stratifed by outcome: death (red), discharge (blue). The dashed vertical line marks the 30-day observation window. The mode of 4–6 days reflect the typical short ICU stay; deaths are concentrated in the first two weeks. **(b)** Risk set size (black) with cumulative discharges (blue) and deaths (red). The widening gap between cumulative rates illustrates that the risk set depletes primarily through discharge, especially in the first week.

was overall younger, and with lower comorbidity burden, consistent with hypothesis that discharge represents a positive prognostic signal.

5.3 Censoring Pattern and Risk Set

Table 6 and Figure 1 show how the risk set composition evolves across the observation period. The risk set declines most rapidly in the first week, driven predominantly by discharge (blue) rather than death (red). By day 30, approximately 3,200 patients remain at risk—a sufficient sample for stable estimation interval-specific intercepts α_t. The front-loading discharge pattern underscores the importance of the sensitivity component. If the sucker patients plausibly, remained longer under observation, the standard analysis would underestimate the hazard in later intervals.

5.4 Analysis

We fit the both models to the MIMIC-III person-period dataset ($R = 509,944$ rows). Model A (standard logistic hazard) serves as the primary analysis, estimated with 4 chains of 2,000 iterations (500 warmup), and `adap_delta` = 0.90. Model B (log-normal frailty, non-centered) used identical settings. To avoid memory overflow from serializing 510K rows we fit the models without the `generated_quantities` block and computed $\log p(y_r \mid \theta^{(s)})$ post-hoc in R from the posterior draws. The sensitivity analysis employed Stan's ADVI (full-rank) across a grid of $\gamma \in \{-1.5, -1.0, -0.5, 0, 0.5, 1.0, 1.5\}$.

5.5 Results

5.5.1 Model A (Standard Logistic Hazard).

Model A converged without difficulty, no divergent transitions, all $\hat{R} < 1.01$, with the minimum bulk ESS of 878 (for emergency admission coefficient). Total clock time was 39 minutes. The posterior means of the 17 covariate effects (log-odds for daily mortality) are reported in Table 7. The strongest risk factor was mechanical ventilation ($\hat{\beta} = 0.96$, 95% CrI [0.91, 1.02]), emergency admission (0.90, [0.81, 1.00]), standardized age (0.76, [0.72, 0.80]), ICD-coded malignancy (0.66, [0.61, 0.72]), GSC impairment (0.61, [0.55, 0.68]), sepsis (0.50, [0.44, 0.56]). These are all very plausible: ventilated patients in emergency admissions face highest daily hazard of ICU death.

Several covariates showed little to no effects. Pneumonia ($\hat{\beta} = -0.35$) and positive blood cultures (-0.29) were associated with lower daily mortality, likely reflecting these diagnoses prompt early aggressive treatment (such as antibiotics), with favorable short-term prognosis. Sex, COPD, vasopressor use, and dialysis shows credible intervals overlapping or near zero, suggesting minimal independent effects on daily mortality risk after adjusting for the other covariates.

5.5.2 Model B (Frailty).

Model B also converged without divergences, though Stan's missing data flag (E-BFMI) flagged the heavy tails in all 4 chains and the ESS for σ_b was only 97 (with $\hat{R} = 1.03$), indicating that the frailty variance was difficult to estimate, providing a consistent result from our initial simulations. The posterior standard deviation was $\hat{\sigma}_b = 1.38$ (95% CrI [1.17, 1.61]), indicating substantial unobserved heterogeneity in patient-level hazard. Including frailty uniformly amplified covariate effects, all 17 coefficients moved awau from zero (Table 7), with the largest increasee for mechanical ventilation (+0.28), sepsis (+0.20), malignancy (+0.19), and emergency admission (+0.17). The correction from the frailty model shows that the standard model conflates unobserved heterogeneity with baseline hazard, biasing fixed effects toward zero.

5.5.3 Interpretation.

The frailty estimate $\hat{\sigma}_b = 1.38$ implies that the 95th-percentile patient (by unobserved frailty) faces a daily hazard roughly $\exp(2 \times 1.38) \approx 16\times$ higher than the 5th-percentile patient, holding observed covariates constant. This is clinically reasonable: unmeasured factors such as pre-morbid functional status, goals-of-care decisions, and unmeasured organ dysfunction create enormous baseline variability. However, the low ESS and E-BFMI warnings confirm that σ_b is only weakly identified from the discrete-time likelihood, reinforcing the motivation for the sensitivity analysis rather than relying exclusively on the frailty model for inference about the censoring mechanism.

The coefficient comparison reveals a consistent pattern: when unobserved heterogeneity is absorbed into a random intercept, the fixed effects become larger in magnitude. This is the classic attenuation bias result for binary outcomes—omitting a random effect biases coefficients in generalized linear models toward zero. In our setting, the practical implication

Table 1: Posterior mean covariate effects (log-odds ratio for daily hazard) under Model A and Model B. Difference column shows the attenuation correction from including frailty.

Covariate	**Model A**	**Model B**	**Difference**
age (std)	0.757	0.921	+0.163
male	−0.092	−0.119	−0.027
emergency	0.902	1.074	+0.172
icd_sepsis	0.502	0.704	+0.202
icd_hf	−0.099	−0.131	−0.031
icd_resp	0.202	0.263	+0.061
icd_aki	0.193	0.245	+0.051
gcs_impaired	0.614	0.718	+0.104
vasopressor	−0.063	−0.035	+0.028
icd_pneumonia	−0.353	−0.493	−0.140
icd_diabetes	−0.139	−0.188	−0.049
icd_liver	0.436	0.515	+0.079
icd_malignancy	0.661	0.852	+0.190
icd_copd	−0.044	−0.068	−0.025
is_ventilated	0.964	1.250	+0.286
received_dialysis	0.046	0.044	−0.002
positive_blood_cx	−0.286	−0.406	−0.120

is that Model A's coefficient estimates represent conservative lower bounds on the true covariate–hazard associations.

Table 2: MCMC diagnostics for covariate effects ($\hat{\beta}$) under both models. $\hat{R}$ values below 1.01 and ESS above 400 indicate reliable convergence. Model B additionally estimates σ_b (frailty SD). All chains: 0 divergent transitions.

	Model A (Standard)					**Model B (Frailty)**				
Parameter	Mean	SD	$\hat{R}$	ESS_b	ESS_t	Mean	SD	$\hat{R}$	ESS_b	ESS_t
σ_b	—	—	—	—	—	1.378	0.133	1.027	97	139
age (std)	0.757	0.0257	1.001	5894	4831	0.921	0.0424	1.013	211	360
male	-0.092	0.0276	1.000	5256	4550	-0.119	0.0365	1.001	4463	4066
emergency	0.902	0.0560	1.004	878	2268	1.074	0.0721	1.007	381	914
icd_sepsis	0.502	0.0346	1.000	5462	3994	0.704	0.0554	1.011	283	752
icd_hf	-0.099	0.0322	1.001	5469	3551	-0.131	0.0417	1.000	5186	4970
icd_resp	0.202	0.0346	1.001	4925	4173	0.263	0.0458	1.001	2284	4286
icd_aki	0.193	0.0318	1.001	6031	4112	0.245	0.0420	1.001	1845	3678
gcs_impaired	0.614	0.0394	1.001	3584	4368	0.718	0.0507	1.004	530	869
vasopressor	-0.063	0.0346	1.001	5472	4179	-0.035	0.0471	1.000	4713	4584
icd_pneumonia	-0.353	0.0343	1.000	5950	4312	-0.493	0.0499	1.005	547	787
icd_diabetes	-0.139	0.0322	1.000	7363	4324	-0.188	0.0414	1.000	3297	4729
icd_liver	0.436	0.0476	1.000	5946	4321	0.515	0.0651	1.001	2273	4281
icd_malignancy	0.661	0.0339	1.000	5393	4400	0.852	0.0540	1.008	294	605
icd_copd	-0.044	0.0382	1.001	7041	4039	-0.068	0.0508	1.000	5386	4469
is_ventilated	0.964	0.0357	1.001	4520	3842	1.250	0.0666	1.015	162	240
received_dialysis	0.046	0.0474	1.001	6644	4486	0.044	0.0621	1.000	5174	5341
positive_blood_cx	-0.286	0.0379	1.000	7050	4530	-0.406	0.0532	1.005	761	1815

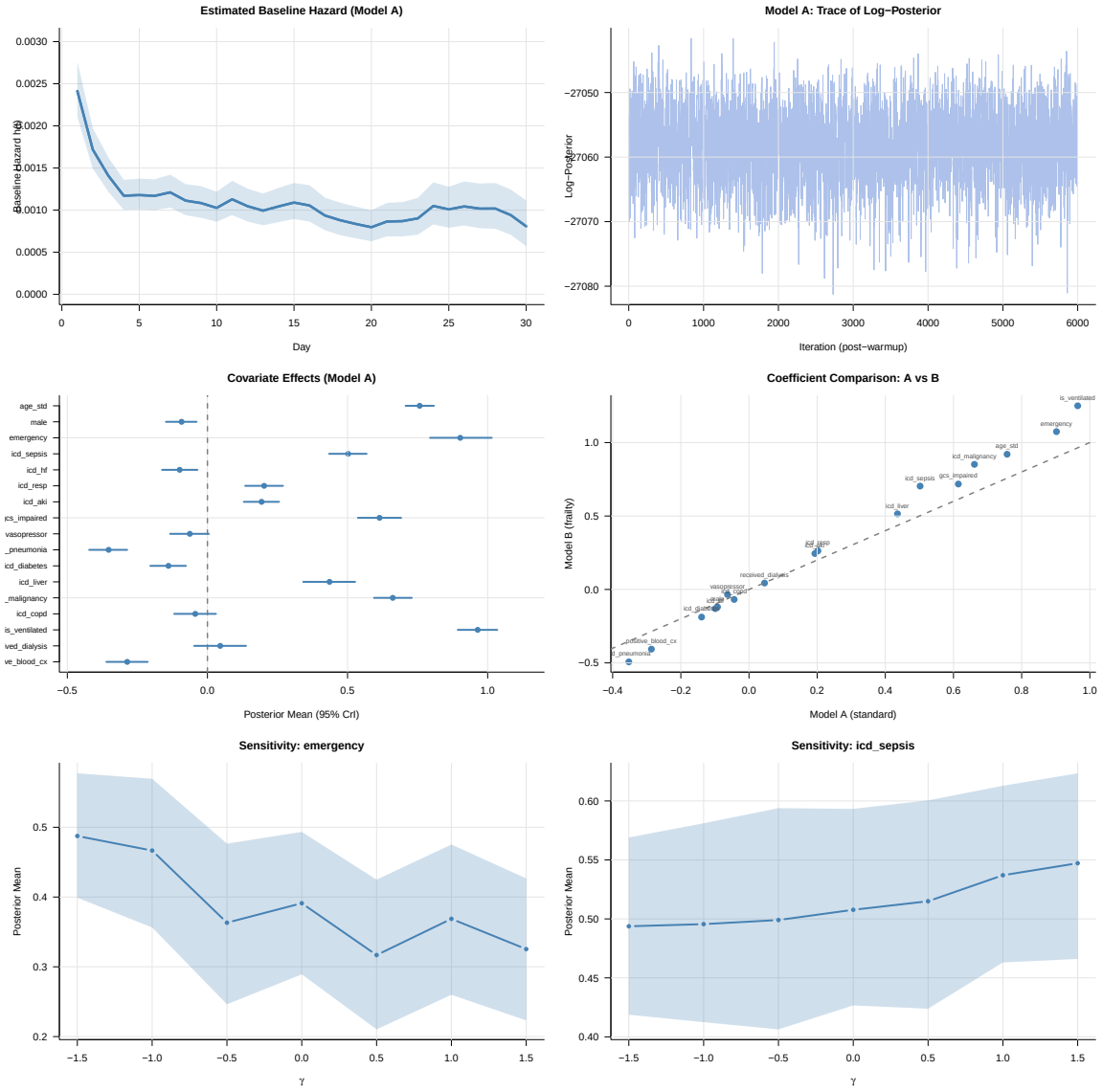

Figure 2: Six-panel overview of the MIMIC-III analysis. **(a)** Estimated baseline hazard $h(t) = \text{logit}^{-1}(\alpha_t)$ with 95% CrI, showing an early peak followed by plateau. **(b)** Trace of log-posterior confirming convergence. **(c)** Forest plot of all 17 covariate effects with 95% CrI. **(d)** Model A vs. B coefficient comparison; all points lie above the diagonal, confirming uniform attenuation correction. **(e–f)** Sensitivity of emergency and sepsis effects to γ; both remain well above zero across the grid.

6. Discussion

[To be completed. Key points to address:]

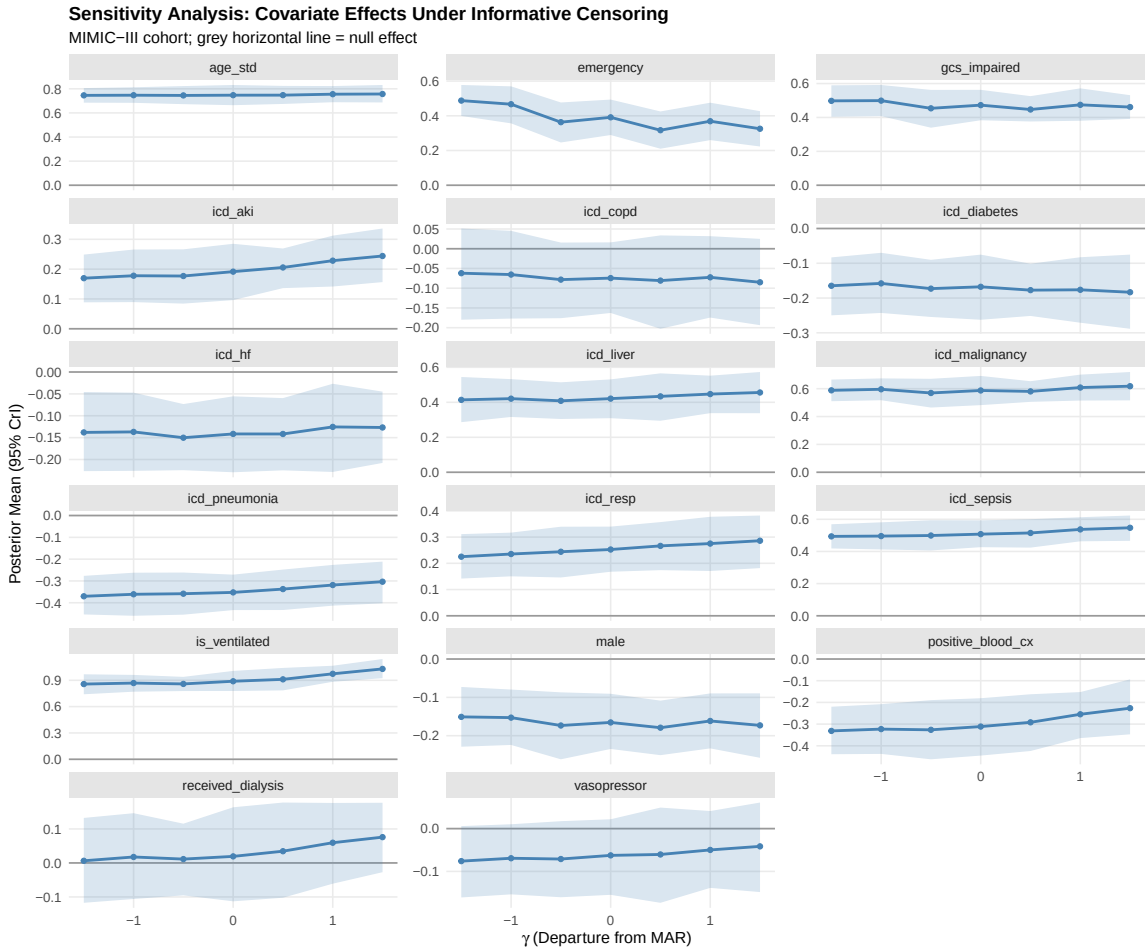

Figure 3: Sensitivity analysis for all 17 covariates. Posterior mean and 95% CrI as a function of γ (departure from MAR). Grey horizontal line indicates null effect. Of 17 covariates, 14 never cross zero for $\gamma \in [-1.5, 1.5]$. Only vasopressor, COPD, and dialysis have tipping points at $\gamma^* = -1.5$.

6.0.1 Summary of Contributions.

6.0.2 Limitations.

6.0.3 Future Work.

Appendix A. Derivation of Biased Hazard under Informative Censoring

At time t the full population of survivors decomposes into the risk set $\mathcal{R}_t = \{i : T_i \geq t, C_i \geq t\}$ and the censored set $\mathcal{C}_t = \{i : T_i \geq t, C_i < t\}$. The true marginal hazard $h_t^* = \mathbb{E}[h_{i,t} \mid T_i \geq t]$ and observed hazard $\hat{h}_t = \mathbb{E}[h_{i,t} \mid i \in \mathcal{R}_t]$ are related by:

$$h_t^* = \Pr(T_i = t \mid T_i \geq t) = \mathbb{E}(h_{i,t} \mid T_i \geq t)$$
$$\hat{h}_t = \Pr(T_i = t \mid T_i \geq t, C_i \geq t) = \mathbb{E}(h_{i,t} \mid T_i \geq t, C_i \geq t) = \mathbb{E}(h_{i,t} \mid i \in \mathcal{R}_t)$$

Where $\lambda_t = \Pr(C_i < T_i \mid T_i \geq t) = |\mathcal{C}_t|/|\mathcal{R}_t|$ as the proportion of survivors (censored at t) relative to the risk set. Then, $\hat{h}_t$ can be expressed as a mixture of the hazards in $\mathcal{R}_t$ and $\mathcal{C}_t$. Let $h_t^c = \mathbb{E}(h_{i,t} \mid i \in \mathcal{C}_t)$ be the mean hazard among those censored by t. Then:

$$h_t^* = (1-\lambda_t)\underbrace{\mathbb{E}(h_{i,t} \mid i \in \mathcal{R}_t)}_{\hat{h}_t} + \lambda_t \underbrace{\mathbb{E}(h_{i,t} \mid i \in \mathcal{C}_t)}_{h_t^c} = (1-\lambda_t)\hat{h}_t + \lambda_t h_t^c \implies \hat{h}_t = \frac{h_t^* - \lambda_t h_t^c}{1-\lambda_t}$$

Defining $\Delta_t := h_t^* - h_t^c$ and rearranging: $\hat{h}_t = h_t^* + \lambda_t \Delta_t/(1-\lambda_t)$. The bias is:

$$\hat{h}_t = \frac{h_t^* - \lambda_t(h_t^* - \Delta_t)}{1-\lambda_t} = \frac{h_t^* - \lambda_t h_t^* + \lambda_t \Delta_t}{1-\lambda_t} = h_t^* + \frac{\lambda_t \Delta_t}{1-\lambda_t}$$
$$\therefore \text{Bias}(\hat{h}_t) = \hat{h}_t - h_t^* = \frac{\lambda_t \Delta_t}{1-\lambda_t} > 0$$

The bias is positive whenever $\lambda_t, \Delta_t > 0$, meaning healthier patients censored earlier and compounds the overestimation of the hazard ($\hat{S}_t = \prod_{s=1}^{t}(1-\hat{h}_s)$). Where as when $\Delta_t < 0$ the bias is negative, meaning sicker patients censored earlier in the case of hospice discharges. The bias would theoretically underestimate the hazard though this is not the norm in the ICU data (See Figure 1).

Appendix B. Semiparametric Estimation Equations under Informative Censoring

Reiterating a key result from Scharfstein et al. (1999) that pertains to this project, censoring induced confounding bias is a failure of the semiparametric estimation equations (score) for $\boldsymbol{\beta}$ when the MAR assumption is violated. When the probability of censoring depends on the unobserved outcome (the failure time T_i), this constitutes a Missing Not at Random (MNAR) or nonignorable censoring mechanism. In this case, the standard score functions used to estimate $\boldsymbol{\beta}$ are biased proportional to the dependence between T_i and C_i given $\mathbf{x}_i$.

Let $\Delta_i = \mathbb{I}(T_i \leq C_i)$ be the observation indicator (0 if censored, 1 if observed). A standard solution under MAR to account for missing data is to use an Inverse Probability Weighting (IPW). If MAR holds, the probability of being observed depends only on $\mathbf{x}_i$, denoted $\pi(\mathbf{x}_i; \alpha = 0) = \Pr(\Delta_i = 1 \mid \mathbf{x}_i)$. When MAR is violated, the true probability of being observed depends on both T_i and $\mathbf{x}_i$, denoted $\pi(T_i, \mathbf{x}_i; \alpha) = \Pr(\Delta_i = 1 \mid T_i, \mathbf{x}_i)$. Again, α is a non-identifiable selection parameter that indexes the severity of the MAR violation.

In standard parametric estimation, the expectation of the score at the true parameter value $\boldsymbol{\beta}_0$ always equals zero. Under informative censoring, however, this is no longer the case. To find the bias, we take the expectation under the MNAR distribution:

$$\text{Bias}(\alpha) = \mathbb{E}\left[\frac{\Delta_i}{\pi(\mathbf{x}_i;0)} S_{\text{full}}(\boldsymbol{\beta}_0)\right] = \mathbb{E}\left[\mathbb{E}\left[\frac{\Delta_i}{\pi(\mathbf{x}_i;0)} S_{\text{full}}(\boldsymbol{\beta}_0)\middle| T_i, \mathbf{x}_i\right]\right] = \mathbb{E}\left[\frac{\pi(T_i,\mathbf{x}_i;\alpha)}{\pi(\mathbf{x}_i;0)} S_{\text{full}}(\boldsymbol{\beta}_0)\right]$$

where $S_{\text{full}}(\boldsymbol{\beta})$ is the full-data score function (i.e., the score function we would have if we observed T_i for all patients). Following Scharfstein et al. (1999), if we specify the conditional hazard of censoring as $\lambda_C(t \mid \mathbf{x}_i, T_i) = \lambda_0(t \mid \mathbf{x}_i)\exp(\alpha T_i)$, where λ_0 is the baseline hazard of censoring, and $\exp(\alpha T_i)$ is the "tilt" caused by unobserved dependence between T_i and C_i. The survival function of censoring at T_i: $\pi(T_i, \mathbf{x}_i; \alpha) = \exp\{-\int_0^{T_i} \lambda_C(u \mid \mathbf{x}_i, T_i)du\} = \exp\{-\int_0^{T_i} \lambda_0(u \mid \mathbf{x}_i)\exp(\alpha T_i)du\} = \exp\{-\exp(\alpha T_i)\Lambda_0(T_i \mid \mathbf{x}_i)\}$, where Λ_0 is the cumulative baseline hazard of censoring. While under MAR, $\pi(\mathbf{x}_i;0) = \exp\{-\Lambda_0(T_i \mid \mathbf{x}_i)\}$. Thus, the ratio of the true to assumed selection probabilities is:

$$\begin{aligned}\frac{\pi(T_i,\mathbf{x}_i;\alpha)}{\pi(\mathbf{x}_i;0)} &= \frac{\exp\{-\exp(\alpha T_i)\Lambda_0(T_i \mid \mathbf{x}_i)\}}{\exp\{-\Lambda_0(T_i \mid \mathbf{x}_i)\}} \\ &\approx \frac{\exp\{-(1+\alpha T_i)\Lambda_0(T_i \mid \mathbf{x}_i)\}}{\exp\{-\Lambda_0(T_i \mid \mathbf{x}_i)\}} \\ &= \exp\{-\alpha T_i \Lambda_0(T_i \mid \mathbf{x}_i)\}\end{aligned}$$

where $e^{\alpha T_i} \approx 1 + \alpha T_i$ in the tilt term on the numerator. Applying a first-order Taylor expansion for small α yields:

$$\frac{\pi(T_i,\mathbf{x}_i;\alpha)}{\pi(\mathbf{x}_i;0)} \approx 1 - \alpha T_i \Lambda_0(T_i \mid \mathbf{x}_i) \tag{18}$$

Substituting this back into the expectation yields the explicit bias approximation:

$$\text{Bias}(\alpha) \approx -\alpha\, \mathbb{E}\Big[\Lambda_0(T_i \mid \mathbf{x}_i)\, T_i\, S_{\text{full}}(\boldsymbol{\beta})\Big] \tag{19}$$

This result explicitly demonstrates the main theoretical consequence of informative censoring: the bias in estimating $\boldsymbol{\beta}$ is strictly driven by α (the severity of the MAR violation/selection) and the covariance (cumulative hazard) between the failure time T_i and the censoring $\Lambda_0(T_i \mid \mathbf{x}_i)$. Because α cannot be identified from the observed data alone, standard estimation fails, motivating the need for alternative approaches such as sensitivity analyses or frailty models (as discussed in Section 3.5) to absorb this unobserved dependence.

Appendix C. Simulation Study

We validate the methodology with synthetic ICU data ($N \in \{300, 500\}$, $K_{\max} = 90$, $\Delta \in \{1, 3, 7, 14\}$) where informative censoring is known: $\text{logit}(h^c_{i,t}) = -2.5 - 0.3\log(t) - 0.8 b_i$, so healthier patients (lower b_i) have higher discharge probability ($\sigma_{b,\text{true}} = 0.6$). Key findings: (i) Model A reliably converges with zero divergences and ESS $>$ 5,000 across all configurations. (ii) Model B's σ_b is weakly identified: ESS $\approx$ 80–150, $\hat{R} \approx$ 1.03–1.08, and

95% CrI spanning [0.016, 1.660] with truth 0.600. LOO-CV (not implemented in main analysis due to computational constraints) never favors Model B. (iii) The sensitivity analysis converges cleanly across $\gamma \in [-2, 2]$ and correctly distinguishes robust from fragile covariate effects. (iv) Model A calibration shows systematic over-prediction consistent with (2).

Appendix D. MIMIC-III Cohort Description

The following tables supplement the cohort description in Section 5.

Table 3: Baseline characteristics of the MIMIC-III analytic cohort within the 30-day observation window, stratified by outcome. *Discharged* denotes patients censored alive before the window closes (the clinically informative censoring mechanism); *Survived* denotes patients still alive at 30 days (administratively censored). Values are n (%) for categorical variables and mean ± SD or median [IQR] for continuous variables.

	Overall (N=57,694)	**Died** (n=5,443)	**Discharged** (n=49,660)	**Survived** (n=2,591)
Demographics				
Age, years (mean ± SD)	55.1 ± 27.3	69.8 ± 17.2	54.2 ± 27.3	40.2 ± 31.1
Male sex	32,248 (55.9%)	2,906 (53.4%)	27,869 (56.1%)	1,473 (56.9%)
Emergency admission	41,580 (72.1%)	5,081 (93.3%)	34,969 (70.4%)	1,530 (59.1%)
Follow-up, days (median [IQR])	5.9 [3.5, 10.6]	4.5 [1.6, 9.8]	5.7 [3.6, 9.6]	42.0 [35.0, 57.3]
Comorbidities (ICD-9)				
Sepsis	6,320 (11.0%)	1,777 (32.6%)	3,878 (7.8%)	665 (25.7%)
Heart failure	13,434 (23.3%)	1,840 (33.8%)	11,036 (22.2%)	558 (21.5%)
Acute kidney injury	11,359 (19.7%)	2,335 (42.9%)	8,176 (16.5%)	848 (32.7%)
Respiratory failure	8,906 (15.4%)	2,460 (45.2%)	5,661 (11.4%)	785 (30.3%)
Pneumonia	7,347 (12.7%)	1,332 (24.5%)	5,322 (10.7%)	693 (26.7%)
Diabetes	13,900 (24.1%)	1,423 (26.1%)	12,078 (24.3%)	399 (15.4%)
Liver disease	2,916 (5.1%)	538 (9.9%)	2,195 (4.4%)	183 (7.1%)
Malignancy	6,730 (11.7%)	1,258 (23.1%)	5,184 (10.4%)	288 (11.1%)
COPD	6,568 (11.4%)	870 (16.0%)	5,508 (11.1%)	190 (7.3%)
Clinical Indicators				
GCS impaired (≤ 12)	30,190 (52.3%)	4,495 (82.6%)	24,358 (49.0%)	1,337 (51.6%)
Mechanical ventilation	14,477 (25.1%)	3,492 (64.2%)	9,240 (18.6%)	1,745 (67.3%)
Vasopressor use	6,836 (11.8%)	1,312 (24.1%)	5,134 (10.3%)	390 (15.1%)
Dialysis	3,166 (5.5%)	596 (10.9%)	2,195 (4.4%)	375 (14.5%)
Positive blood culture	5,448 (9.4%)	1,049 (19.3%)	3,546 (7.1%)	853 (32.9%)

Table 4: Summary of the discrete-time data structure within the 30-day observation window ($\Delta = 1$ days, $K = 30$ intervals). Clinically censored patients are those discharged alive before 30 days; administratively censored patients survived the full window.

Unique admissions (N)	57,694
Outcome within observation window	
Died within 30 days	5,443 (9.4%)
Discharged alive $<$ 30 days	49,660 (86.1%)
Survived $\geq$ 30 days	2,591 (4.5%)
Discrete-time structure	
Discretization width (Δ)	1 days
Maximum intervals (K)	30
Observation window	30 days
Static covariates (P)	17
Person-period rows ($\mathcal{R}$)	509,944
Event rows ($y_{i,t} = 1$)	5,443 (1.067% of $\mathcal{R}$)
Follow-up time, days	
Overall: median [IQR]	5.9 [3.5, 10.6]
Decedents: median [IQR]	4.5 [1.6, 9.8]
Discharged: median [IQR]	5.7 [3.6, 9.6]
Survived $\geq$ 30 days	2,591 patients

Table 5: Covariates in the design matrix $\mathbf{X}$ ($P = 17$). The first nine covariates (above the rule) correspond to the simulation study; the remaining eight are additional clinical predictors. All variables except age are binary (0/1). ICD-9 codes shown in parentheses.

	Variable	**Description**	**Prevalence**
1	`age_std`	Age (standardized to $\mu = 0$, $\sigma = 1$)	55.1 ± 27.3 yr
2	`male`	Male sex	55.9%
3	`emergency`	Emergency admission	72.1%
4	`icd_sepsis`	Sepsis (995.91/92, 038.x)	11.0%
5	`icd_hf`	Heart failure (428.x)	23.3%
6	`icd_resp`	Respiratory failure (518.8x)	15.4%
7	`icd_aki`	Acute kidney injury (584.x)	19.7%
8	`gcs_impaired`	GCS ≤ 12	52.3%
9	`vasopressor`	Any vasopressor	11.8%
10	`icd_pneumonia`	Pneumonia (480–486)	12.7%
11	`icd_diabetes`	Diabetes (250.x)	24.1%
12	`icd_liver`	Liver disease (571.x)	5.1%
13	`icd_malignancy`	Malignancy (140–208)	11.7%
14	`icd_copd`	COPD (490–492, 494, 496)	11.4%
15	`is_ventilated`	Mechanical ventilation (proc 96.7x)	25.1%
16	`received_dialysis`	Hemodialysis (proc 39.95)	5.5%
17	`positive_blood_cx`	Positive blood culture	9.4%

Table 6: Risk set composition across discrete-time intervals ($\Delta = 1$ days). *Censored* here denotes patients discharged alive during each interval. The censoring proportion among exits quantifies the rate at which the informative censoring mechanism depletes the risk set—the empirical pattern interrogated by the sensitivity parameter γ in Section 3.6.

Interval	Days	At Risk	Events	Discharged	% Disch.
1	1–1	57,694	994	1,484	59.9%
2	2–2	55,216	659	3,274	83.2%
3	3–3	51,283	515	5,377	91.3%
4	4–4	45,391	387	5,195	93.1%
5	5–5	39,809	363	6,148	94.4%
10	10–10	17,486	161	1,998	92.5%
15	15–15	9,738	109	887	89.1%
20	20–20	6,039	41	446	91.6%
25	25–25	4,001	34	266	88.7%
30	30–30	2,765	12	2,753	99.6%
Total	1–30	—	5,443	52,251	—

Table 7: Posterior mean covariate effects (log-odds ratio for daily hazard) under Model A and Model B. Difference column shows the attenuation correction from including frailty.

Covariate	**Model A**	**Model B**	**Difference**
age (std)	0.757	0.921	+0.163
male	−0.092	−0.119	−0.027
emergency	0.902	1.074	+0.172
icd_sepsis	0.502	0.704	+0.202
icd_hf	−0.099	−0.131	−0.031
icd_resp	0.202	0.263	+0.061
icd_aki	0.193	0.245	+0.051
gcs_impaired	0.614	0.718	+0.104
vasopressor	−0.063	−0.035	+0.028
icd_pneumonia	−0.353	−0.493	−0.140
icd_diabetes	−0.139	−0.188	−0.049
icd_liver	0.436	0.515	+0.079
icd_malignancy	0.661	0.852	+0.190
icd_copd	−0.044	−0.068	−0.025
is_ventilated	0.964	1.250	+0.286
received_dialysis	0.046	0.044	−0.002
positive_blood_cx	−0.286	−0.406	−0.120

References

Paul D. Allison. Discrete-time methods for the analysis of event histories. *Sociological Methodology*, 13:61–98, 1982. doi: 10.2307/270718.

Michael J. Daniels and Joseph W. Hogan. *Missing Data in Longitudinal Studies: Strategies for Bayesian Modeling and Sensitivity Analysis*. Chapman & Hall/CRC, Boca Raton, FL, 2008. ISBN 978-1584886099.

Andrew Gelman. Prior distributions for variance parameters in hierarchical models (comment on article by Browne and Draper). *Bayesian Analysis*, 1(3):515–534, 2006. doi: 10.1214/06-BA117A.

Matthew D. Hoffman and Andrew Gelman. The No-U-Turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15 (47):1593–1623, 2014. URL `http://jmlr.org/papers/v15/hoffman14a.html`.

Alistair Johnson, Tom Pollard, and Roger Mark. MIMIC-III Clinical Database. *PhysioNet*, September 2016. doi: 10.13026/C2XW26. URL `https://doi.org/10.13026/C2XW26`. Version 1.4.

John D. Kalbfleisch and Ross L. Prentice. *The Statistical Analysis of Failure Time Data*. John Wiley & Sons, Hoboken, N.J., 2nd edition, 2003. ISBN 978-0-471-36357-6.

Table 8: MCMC diagnostics for covariate effects ($\hat{\beta}$) under both models. $\hat{R}$ values below 1.01 and ESS above 400 indicate reliable convergence. Model B additionally estimates σ_b (frailty SD). All chains: 0 divergent transitions.

	Model A (Standard)					**Model B (Frailty)**				
Parameter	Mean	SD	$\hat{R}$	ESS_b	ESS_t	Mean	SD	$\hat{R}$	ESS_b	ESS_t
σ_b	—	—	—	—	—	1.378	0.133	1.027	97	139
age (std)	0.757	0.0257	1.001	5894	4831	0.921	0.0424	1.013	211	360
male	-0.092	0.0276	1.000	5256	4550	-0.119	0.0365	1.001	4463	4066
emergency	0.902	0.0560	1.004	878	2268	1.074	0.0721	1.007	381	914
icd_sepsis	0.502	0.0346	1.000	5462	3994	0.704	0.0554	1.011	283	752
icd_hf	-0.099	0.0322	1.001	5469	3551	-0.131	0.0417	1.000	5186	4970
icd_resp	0.202	0.0346	1.001	4925	4173	0.263	0.0458	1.001	2284	4286
icd_aki	0.193	0.0318	1.001	6031	4112	0.245	0.0420	1.001	1845	3678
gcs_impaired	0.614	0.0394	1.001	3584	4368	0.718	0.0507	1.004	530	869
vasopressor	-0.063	0.0346	1.001	5472	4179	-0.035	0.0471	1.000	4713	4584
icd_pneumonia	-0.353	0.0343	1.000	5950	4312	-0.493	0.0499	1.005	547	787
icd_diabetes	-0.139	0.0322	1.000	7363	4324	-0.188	0.0414	1.000	3297	4729
icd_liver	0.436	0.0476	1.000	5946	4321	0.515	0.0651	1.001	2273	4281
icd_malignancy	0.661	0.0339	1.000	5393	4400	0.852	0.0540	1.008	294	605
icd_copd	-0.044	0.0382	1.001	7041	4039	-0.068	0.0508	1.000	5386	4469
is_ventilated	0.964	0.0357	1.001	4520	3842	1.250	0.0666	1.015	162	240
received_dialysis	0.046	0.0474	1.001	6644	4486	0.044	0.0621	1.000	5174	5341
positive_blood_cx	-0.286	0.0379	1.000	7050	4530	-0.406	0.0532	1.005	761	1815

Radford M. Neal. Slice sampling. *The Annals of Statistics*, 31(3):705–767, 2003. doi: 10.1214/aos/1056562461. Section 6 introduces the "funnel" distribution illustrating hierarchical sampling difficulties.

Nicholas G. Polson, James G. Scott, and Jesse Windle. Bayesian inference for logistic models using pólya–gamma latent variables. *Journal of the American Statistical Association*, 108 (504):1339–1349, 2013. doi: 10.1080/01621459.2013.829001. URL `https://doi.org/10.1080/01621459.2013.829001`.

Donald B. Rubin. Inference and missing data. *Biometrika*, 63(3):581–592, 1976. doi: 10.1093/biomet/63.3.581.

Daniel O. Scharfstein, Andrea Rotnitzky, and James M. Robins. Adjusting for nonignorable drop-out using semiparametric nonresponse models. *Journal of the American Statistical Association*, 94(448):1096–1120, 1999. doi: 10.1080/01621459.1999.10473862.

Daniel P. Simpson, Håvard Rue, Andrea Riebler, Thiago G. Martins, and Sigrunn H. Sørbye. Penalising model component complexity: A principled, practical approach to constructing priors. *Statistical Science*, 32(1):1–28, 2017. doi: 10.1214/16-STS576.

Stan Development Team. *CmdStan User's Guide, Version 2.34*, 2024a. URL `https://mc-stan.org/docs/cmdstan-guide/variational_config.html#variational-algorithms`.

Stan Development Team. Stan modeling language users guide and reference manual, version 2.35, 2024b. URL `https://mc-stan.org`.

Anastasios Tsiatis. A nonidentifiability aspect of the problem of competing risks. *Proceedings of the National Academy of Sciences*, 72(1):20–22, 1975. doi: 10.1073/pnas.72.1.20.